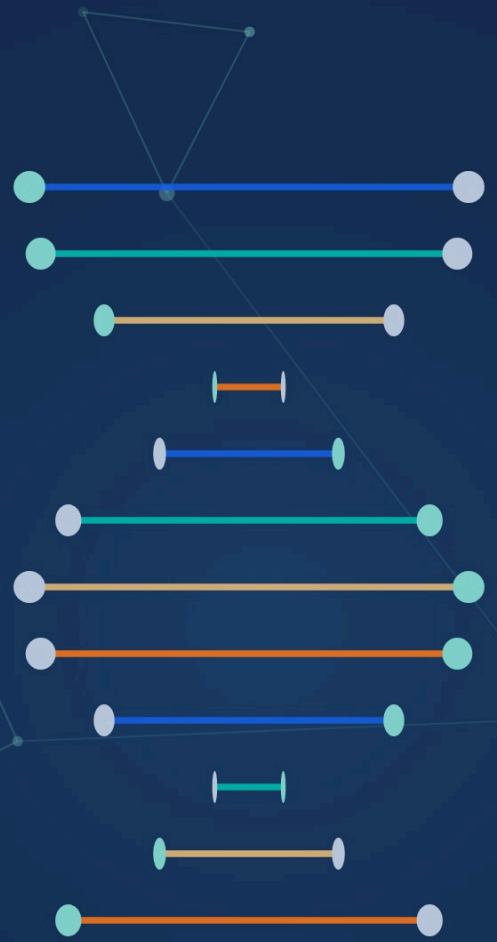
Cyberbiosecurity ÇORUM 2026

DAY 2 · TUESDAY 3 NOVEMBER 2026 · 14:30–15:15

Hands-on: Apply it.

Build a CSF 2.0 profile for a biotechnology laboratory in 45 minutes

Prof. Dr. Ahmet Altun
Sivas Cumhuriyet University · Faculty of Medicine
Cyberbiosecurity Training Workshop · Hitit University, Çorum



HANDS-ON GROUP EXERCISE

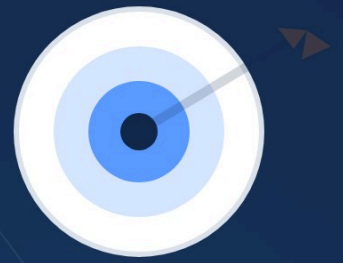
APPLY IT.

NIST CSF 2.0 in a biotechnology laboratory

PROF. DR. AHMET ALTUN

YOUR MISSION

You are HelixLab's new cyberbiosecurity working group.



DELIVER 1

Crown jewels and scope

DELIVER 2

Current and Target Profile for 12 categories

DELIVER 3

Top three actions, with owners

Presented in **three minutes per group** at 15:30.

FACT SHEET · HELIXLAB (FICTIONAL)

A university biotechnology laboratory, BSL-2



PEOPLE

6 staff · 18 graduate students · 2 client technicians with badge access

CLIENTS

3 biotech companies send samples and receive sequencing data

NETWORK

One campus network for everything; lab Wi-Fi password on the door

SYSTEMS

NGS sequencer

Windows 10 control PC; vendor cloud upload; vendor remote support tool

2 × qPCR systems

Results exported by USB stick to the analysis PC

4 × −80 °C freezers

Wi-Fi alarm module sends SMS to the lab head

LIMS

Cloud service; one shared admin account

Analysis server

Linux; 18 student accounts; no central logs

AI tools

Chat assistant for SOP drafts; image classifier for cell counts

THREE THINGS THAT HAPPENED THIS MONTH

1

FRIDAY 18:30

An email from the “sequencer vendor” asks the lab to install an urgent firmware update via a download link before Monday.

2

LAST WEEK

Freezer alarm SMS messages stopped arriving three days ago. Nobody noticed until a sample check.

3

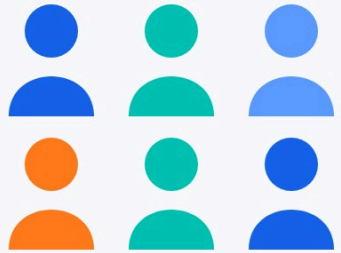
YESTERDAY

A student pasted a client’s sample sheet, with project codes and donor ages, into a public AI chatbot to reformat it.

Use them as evidence when you rate the Current Profile. **Each one points to at least two categories.**

TAKE A ROLE

Six seats at the table. Argue from yours.



Lab head

Chairs; owns the final decisions

Biosafety officer

Samples, containment, cold chain

IT / network

Systems, accounts, logs, backups

Data steward

Client data, KVKK, AI tool use

Vendor manager

Contracts, remote access, procurement

Rapporteur

Records ratings; presents at 15:30

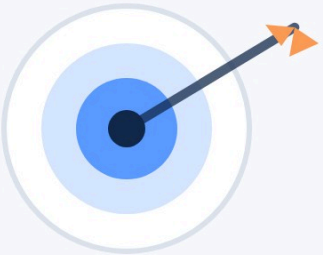
45 MINUTES · FIVE PHASES

1	Scope & crown jewels	8 min
2	Rate the Current Profile	12 min
3	Set Target & find gaps	10 min
4	Top-3 action plan	10 min
5	Prepare the 3-minute pitch	5 min



PHASE 1 · 8 MINUTES · GV.OC, ID.AM

Scope it, then name the crown jewels



ANSWER TOGETHER

- Which systems are in scope? Which are not?
- Which laws and contracts apply? (Law 7545, KVKK, client contracts)
- Who would be harmed if HelixLab were down for two weeks?

CROWN JEWELS · PICK THREE

Lost — What must never be lost?
Availability

Leaked — What must never leak?
Confidentiality

Altered — What must never be altered without anyone noticing?
Integrity

PHASES 2–3 · RATING GUIDE

Rate each category from 1 to 4

1

Partial

Nobody owns it; it happens if someone remembers

2

Risk informed

People know it matters; it is not written policy

3

Repeatable

Written, approved, done the same way every time

4

Adaptive

Measured and improved from lessons learned

`GV.RR` `GV.PO` `GV.SC` `ID.AM` `ID.RA` `PR.AA` `PR.AT` `PR.DS` `PR.PS` `DE.CM` `RS.MA` `RC.RP`

Twelve categories chosen from the 22 for this lab. **Be honest about Current; be realistic about Target.** Most targets land on 2 or 3.

EXAMPLE WORKSHEET · HELIXLAB

One row per category, one radar per group

Code	Category	Current	Target
GV.RR	Roles, responsibilities & authorities	●	● ● ●
GV.PO	Policy	●	● ● ●
GV.SC	Supply chain risk management	●	● ● ●
ID.AM	Asset management	● ●	● ● ●
ID.RA	Risk assessment	●	● ● ●
PR.AA	Identity, authentication & access	●	● ● ●
PR.AT	Awareness & training	● ●	● ● ●
PR.DS	Data security	●	● ● ●
PR.PS	Platform security	●	● ●
DE.CM	Continuous monitoring	●	● ● ●
RS.MA	Incident management	●	● ● ●
RC.RP	Recovery plan execution	● ●	● ● ●



INTERACTIVE COMPANION

Rate live, compare groups, keep the clock.



Phase timer

Counts down each of the five phases and moves on by itself

Profile builder

Example and Group 1–6 tabs; click tiers; radar and top-three gaps update

Action plan

Three rows: action, CSF code, owner, date, cost

`claude.ai/artifact/1LZkxgu56oodvpnb2BBqLr` open on a second tab

PHASE 4 · 10 MINUTES · THE ACTION PLAN

Three actions. Each closes a named gap.

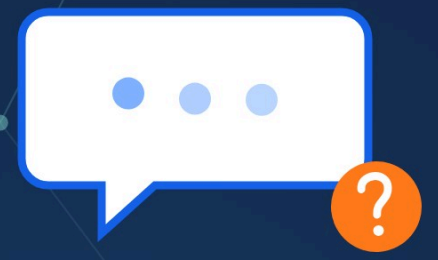


#	Action	CSF	Owner	By when	Cost
e.g.	Route all vendor remote support through one jump host with MFA and a ticket	PR.AA · GV.SC	IT / network	6 weeks	M
1					
2					
3					

FACILITATOR Push back on vague actions. “Improve awareness” of what, for whom, by when?

15:30 · THREE MINUTES PER GROUP

The 180-second pitch



30″ Our crown jewels

45″ Our biggest gap, and the evidence

75″ Our three actions and owners

30″ One question for the room

Bar widths are to scale. The action plan gets the most time.

FACILITATOR DEBRIEF · AFTER THE PRESENTATIONS

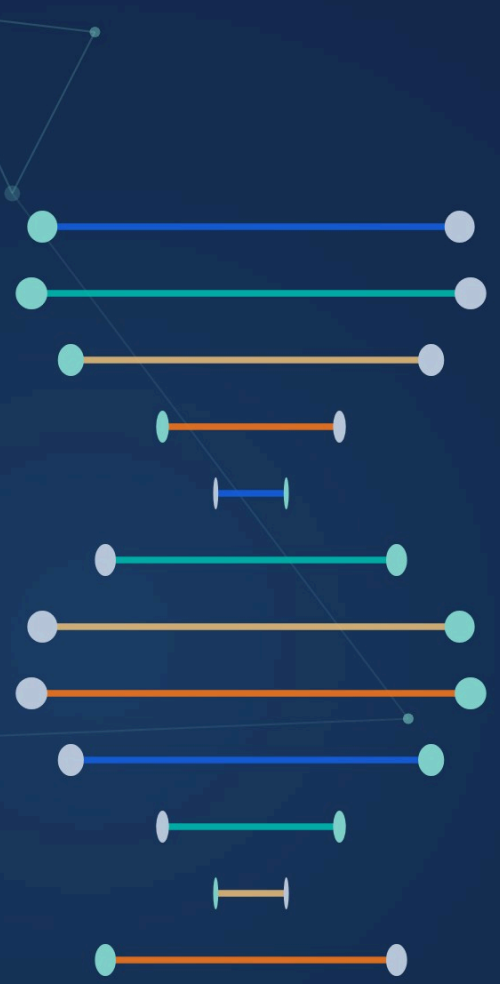
What the three events were pointing at



1	Fake vendor email	GV.SC	PR.AT	PR.AA
2	Freezer alarm went silent	DE.CM	ID.AM	RS.MA
3	Client data in a public chatbot	PR.DS	GV.PO	PR.AT

PATTERN Most groups find their largest gap in GV.SC or DE.CM. Neither needs a big budget; both need an owner.

Your profile is a draft.
Your lab is real.

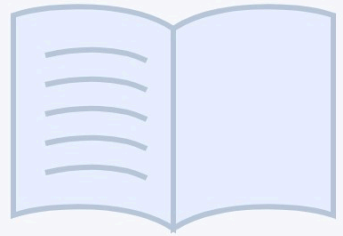


15:30 · Group presentations & discussion

Closing · evaluation · certificates

SOURCES

References and further reading



1. National Institute of Standards and Technology. *The NIST Cybersecurity Framework (CSF) 2.0*. NIST CSWP 29; 2024. doi:10.6028/NIST.CSWP.29
2. National Institute of Standards and Technology. *NIST Cybersecurity Framework 2.0: Organizational Profiles Quick-Start Guide*. NIST SP 1301; 2024. doi:10.6028/NIST.SP.1301
3. NIST. *CSF 2.0 Organizational Profile Template*; 2024. csrc.nist.gov
4. 7545 sayılı Siber Güvenlik Kanunu. Resmî Gazete 32846; 19 Mar 2025. · 6698 sayılı Kişisel Verilerin Korunması Kanunu.
5. HelixLab is a fictional teaching scenario written for this workshop.

QUESTIONS & DISCUSSION

Thank you.

Group presentations follow at 15:30



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